

Summer Internship Report

Interactions of IDPs with Neuronal Model Membranes

Summarized by

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Introduction

Intrinsically disordered proteins (IDPs) play critical roles in physiological and pathological processes, particularly in the central nervous system. Their conformational flexibility allows dynamic interactions with diverse molecular partners, including lipid membranes. This internship focused on understanding how IDPs such as α -synuclein and A β 42 interact with neuronal model membranes, specifically large unilamellar vesicles (LUVs) and mycobacterial extracellular vesicles (MEVs).

The work spanned three interconnected threads. First, MEVs were isolated from *Mycobacterium tuberculosis* cultures and characterized by dynamic light scattering (DLS) and BCA assay; their ability to cross the blood–brain barrier (BBB) was probed by immunostaining of mouse brain tissue. Second, the ESAT-6 protein was expressed in BL21 *E. coli* via IPTG-induced transformation of pESAT plasmids, and expression was confirmed by SDS-PAGE. Third, LUVs of defined lipid composition were prepared by extrusion and sonication for membrane anisotropy studies with α -synuclein, A β 42, and τ protein.

Together, these experiments provided hands-on exposure to protein expression, vesicle isolation, membrane biophysics, and quantitative assay methods relevant to neurodegeneration research.

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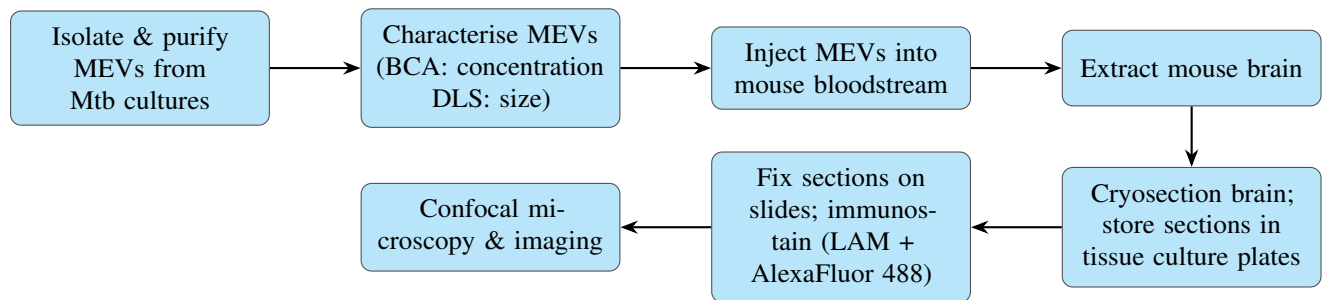
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1 Observing Whether MEVs Cross the Blood–Brain Barrier

1.1 Objective

To determine whether mycobacterial extracellular vesicles (MEVs) can traverse the blood–brain barrier in a mouse model, providing a basis for studying MEV interactions with brain cell types and their pathological implications in neurodegenerative disease.

1.2 Experimental Scheme



Note: Steps were either assisted hands-on or observed.

Figure 1: Workflow for the MEV blood–brain barrier crossing experiment.

MEVs were isolated from *Mycobacterium tuberculosis* H37Ra cultures by sequential centrifugation, 200 nm filtration, tangential flow filtration (TFF), and ultracentrifugation. Purified MEVs were injected intravenously into mice. Brain tissues were subsequently extracted, cryosectioned, and immunostained; confocal microscopy quantified fluorescence intensity as a proxy for MEV penetration into brain parenchyma.

2 Confirming Induction of pESAT Plasmids

2.1 Objective

To confirm that the ESAT-6 protein-producing gene carried on pESAT plasmids is successfully induced in BL21 *E. coli*.

2.2 Experimental Scheme

Competent DH5 α cells were prepared and transformed with pESAT plasmids. Plasmid concentration was estimated by NanoDrop spectrophotometry. BL21 cells were then transformed with the isolated plasmids. Two cultures were prepared: one induced with IPTG and one uninduced (control). Both were lysed by sonication and the lysates were run on SDS-PAGE. Appearance of a band at the expected molecular weight in the induced lane confirmed successful expression.

3 Immunostaining Mouse Brain Tissues

Mice were injected with *Mycobacterium tuberculosis*. MEVs produced by the bacteria enter the bloodstream, and their presence in brain tissue — detected by immunostaining — indicates BBB crossing. Brain tissues were cryosectioned, fixed on slides, and stained; confocal microscopy resolved fluorescence from secondary antibody bound to MEV-associated proteins.

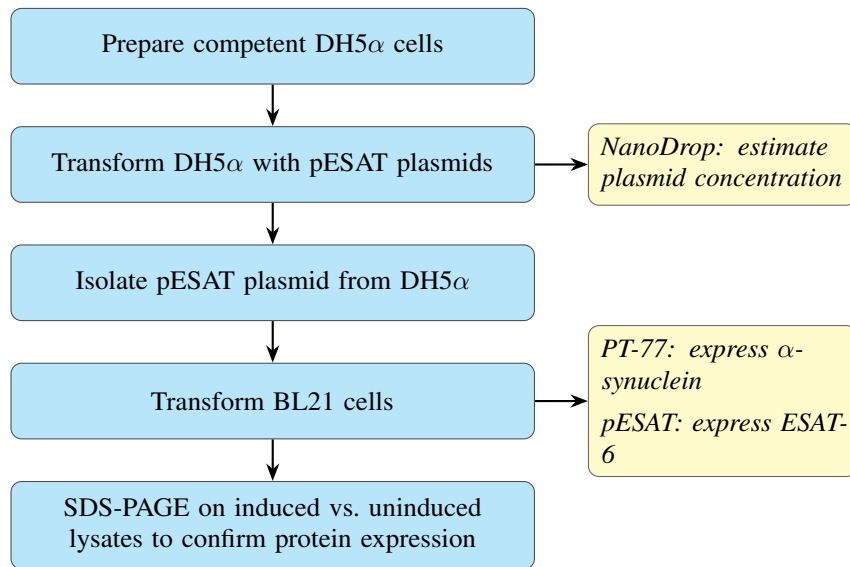


Figure 2: Workflow for confirming ESAT-6 and α -synuclein induction in BL21 *E. coli*.

3.1 Cryosectioning

1. **Fixation.** The dissected brain is immersed in 4% paraformaldehyde (PFA), which crosslinks proteins and preserves tissue morphology.
2. **Cryoprotection.** The fixed brain is sequentially incubated in 15%, 20%, and 30% sucrose (24 h each). Stepwise sucrose infiltration replaces tissue water and prevents ice crystal formation during freezing.
3. **OCT embedding.** Optimal Cutting Temperature (OCT) compound is dispensed onto a disc and frozen to form a stable base for mounting.
4. **Sectioning.** The mounted brain is sectioned at 18–30 μ m on a cryostat, producing slices thin enough for downstream imaging.
5. **Storage.** Sections are transferred to glass slides and stored at -20°C , or placed in $1\times$ PBS (pH 7.4) at 4°C for short-term culture use.



Figure 3: Cryosectioning setup and representative tissue sections.

3.2 Tissue Fixation and Immunostaining

1. Wash slide-mounted tissue $3\times$ with $1\times$ PBS (pH 7.4), 5 min intervals.

2. Fix with 4% PFA to immobilize antigens; wash 3× with PBS.
3. Block with 5% BSA and 5% PBST for 45 min. BSA prevents non-specific antibody binding; PBST permeabilizes membranes and reduces background.
4. Wash 3× with PBS. Apply primary antibody LAM (1:100 in 1× PBST); incubate overnight at 4°C.
5. Wash 3× with PBS. Apply AlexaFluor 488-conjugated anti-rabbit secondary antibody (1:1000 in 1× PBS); incubate 2 h at room temperature. The secondary antibody amplifies signal and emits green fluorescence detectable by confocal microscopy.

4 Transformation of BL21 *E. coli*

BL21 *E. coli* was chosen as the expression host because it lacks Lon and OmpT proteases (preventing recombinant protein degradation), carries T7 RNA polymerase (DE3 lysogen, enabling high-level expression of T7-promoter-driven genes upon IPTG induction), and has reduced restriction systems allowing stable maintenance of foreign plasmids. The expression vector used was PT-77 (ampicillin resistant), carrying the α -synuclein gene.

Table 1: Advantages of the BL21 *E. coli* expression strain.

Feature	Benefit
Deficient in Lon and OmpT proteases	Prevents degradation of recombinant proteins
T7 RNA polymerase (DE3 lysogen)	High-level expression of T7-promoter genes via IPTG
Lacks <i>hsd</i> restriction system	Stable maintenance of foreign plasmid DNA
Reduced endogenous protease/RNase	Improved protein yield and purity
High-density growth capability	Suitable for large-scale production

4.1 Competent Cell Preparation

Three solutions are used: Solution I ($\text{MgCl}_2 + \text{CaCl}_2$ in ddH₂O), Solution II (higher-concentration CaCl_2), and Solution III (glycerol in ddH₂O, cryoprotectant).

1. Inoculate a single colony in 5 mL LB broth overnight at 200 rpm (primary culture). Inoculate 1 mL into 100 mL LB broth; grow to $\text{OD}_{600} = 0.6$ (mid-log phase).
2. Chill on ice for 10 min. Centrifuge at $2700 \times g$, 4°C, 10 min; discard supernatant.
3. Resuspend pellet in 20 mL ice-cold Solution I; incubate on ice 30 min. Divalent Mg^{2+} and Ca^{2+} permeabilize the membrane and facilitate DNA binding.
4. Centrifuge at 2700 rcf, 4°C, 30 min; discard supernatant. Resuspend in 3.2 mL ice-cold Solution II; stir on ice 30 min. Higher Ca^{2+} concentration further permeabilizes the membrane.
5. Add 1 mL Solution III (glycerol) and mix gently. Aliquot 100 μL each on ice; store immediately at -80°C .

4.2 Transformation

1. Mix 100–500 ng plasmid with 100 μL competent cells; incubate on ice 30 min.
2. Heat shock at 42°C for 45 s (creates transient membrane pores for DNA entry). Immediately chill on ice for 2 min (reseals pores).
3. Add 1 mL LB broth; shake 1 h at 37°C for recovery and ampicillin resistance gene expression.
4. Centrifuge at 5000 rpm, 5 min, room temperature. Discard 900 μL supernatant; resuspend pellet in remaining 100 μL .
5. Plate on LB agar + ampicillin (50 $\mu\text{g}/\text{mL}$) plates. Check for colonies after 16 h. Only transformed cells survive ampicillin selection.

5 HT-22 Cell Maintenance

HT-22 are immortalized mouse hippocampal neuronal cells, highly sensitive to glutamate-induced excitotoxicity. They serve as a neuronal model for studying membrane–protein interactions.

5.1 Media Preparation

RPMI media was filtered through a 0.2 µm membrane into an autoclaved bottle. Heat-inactivated FBS (65°C, 30–40 min; not exceeding 60 min to prevent degradation) was added at 10%, along with AntiAnti (1%) and HEPES (2%). The supplemented media was aliquoted into 50 mL falcons and stored at 4°C.

5.2 Cell Passaging

1. Aspirate DMEM+FBS from T25 flask. Add 1 mL trypsin; incubate at 37°C for 4 min until colour change. Dislodge cells by jerking flask.
2. Add 2 mL DMEM+FBS (90:10) to neutralize trypsin. Centrifuge at 2000 rpm, 5 min.
3. Discard supernatant; resuspend pellet in fresh media. Transfer to T75 flask with figure-of-eight agitation for even distribution.

5.3 Cell Seeding and LysoTracker Staining

1. **PDL coating.** Coat coverslips with Poly-D-Lysine (PDL, 0.01 mg/mL) in confocal dishes; incubate at 37°C, 5% CO₂, 1 h. Wash vigorously with MilliQ (≥ 90 times; PDL is toxic to cells). Air dry 1 h in open hood.
2. **Seeding.** Detach cells with trypsin, centrifuge, resuspend in DMEM+FBS. Seed at $\sim 5 \times 10^4$ cells/well in 24-well plates. Rock gently; incubate at 37°C, 5% CO₂, until 60–70% confluency.
3. **LysoTracker staining.** Prepare LysoTracker solution at 50–75 nM in pre-warmed medium. Remove old medium; add staining medium; incubate at 37°C, 30 min. Wash 2–3 $\times$ with PBS.
4. **Fixation.** Fix with 4% PFA, 10 min, room temperature. Wash 3 $\times$ with PBS (5 min each).

5.4 Cryopreservation

Cryomedia: 90% FBS + 10% DMSO (e.g. 2700 µL FBS + 300 µL DMSO per 3 mL). Filter-sterilize before use. DMSO prevents ice crystal formation during freezing; it is toxic at high concentrations and should be handled carefully.

6 Transfection

Transfection is the introduction of nucleic acids into eukaryotic cells by non-viral methods. Here, lipofectamine-mediated transfection was used to deliver fluorescently tagged constructs into HT-22 cells for imaging.

6.1 Principle and Components

Lipofectamine LTX forms lipid vesicles (lipoplexes) with DNA. Plus Reagent enhances DNA condensation and complexing efficiency. Opti-MEM (serum-free DMEM) was used as the transfection medium; the absence of serum stresses cells and promotes nucleic acid uptake.

DNA constructs used: RFP-KDEL (1.4 µL), BFP (9.9 µL), VAPA (2.52 µL), S2 (1.05 µL), P1 (4.56 µL).

6.2 Procedure

1. Prepare Solution A: 250 µL Opti-MEM + 2 µL Plus Reagent + DNA (each construct in a separate tube; one tube with no DNA serves as control).
2. Prepare Solution B: 250 µL Opti-MEM + 2 µL Lipofectamine LTX.
3. Combine A into B; incubate in hood 20 min to allow lipoplex formation.
4. Add lipoplex-medium mixture to cells (maintained in Opti-MEM). Incubate at 37°C, 5% CO₂, for 6 h.

The control tube (no DNA) establishes background effects of lipofectamine alone. DNA–lipid complexes fuse with the cell membrane and deliver DNA cargo intracellularly.

7 Biophysical Methods

7.1 Dynamic Light Scattering (DLS)

DLS measures size distributions of particles in solution by detecting temporal fluctuations in laser light scattered at a fixed angle θ . Smaller particles diffuse faster (Brownian motion) and produce higher-frequency fluctuations; larger particles produce slower fluctuations. The position of the dominant peak in the intensity-weighted size distribution gives the most abundant particle size; peak width reflects polydispersity.

DLS was used to characterize the size of LUVs prepared for anisotropy studies. 20 μL lipid was diluted into 980 μL PBS, mixed, and measured.

7.2 BCA Protein Assay

The BCA assay estimates protein concentration by the reduction of Cu^{2+} to Cu^+ by peptide bonds under alkaline conditions, followed by chelation of Cu^+ by bicinchoninic acid to form a purple complex absorbing at 562 nm. Absorbance is compared to a BSA standard curve (0.125–2.0 mg/mL) to determine MEV protein concentration.

7.2.1 Serial Dilution of MEV Sample

Tube	Conc. (mg/mL)	MEV (μL)	PBS (μL)
A	2.000	20	0
B	1.000	10	10
C	0.500	10	10
D	0.250	10	10
E	0.125	10	10

Each dilution uses the previous tube as source. Mix thoroughly after each step.

7.2.2 Assay Protocol

1. Prepare BCA Working Reagent (WR): mix Reagent A and Reagent B in 50:1 ratio; use fresh.
2. Add 20 μL of each MEV dilution to a tube or well; add 200 μL WR. Mix gently.
3. Incubate at 37°C for 30 min; cool to room temperature.
4. Measure absorbance at 562 nm. Subtract blank (PBS + WR). Run standards in parallel; use standard curve to convert absorbance to concentration. Perform all samples and standards in triplicate.

7.3 Bligh–Dyer Lipid Extraction

The Bligh–Dyer method extracts total lipids from biological samples using a monophasic chloroform/methanol/water system that transitions to a biphasic system upon addition of more chloroform and water. Lipids partition into the lower organic phase.

7.3.1 Reagents (per 1 g wet sample)

7.3.2 Procedure

1. Centrifuge impure lipid emulsion at $4700 \times g$, 7 min; wash pellet with 1% PBS; centrifuge again.
2. Add water:chloroform:methanol (0.8:1:2 v/v/v); for 10 mL total: 2.1 mL water, 2.65 mL chloroform, 5.25 mL methanol. Sonicate 2 min.

Table 2: Bligh–Dyer reagent volumes per 1 g wet sample.

Reagent	Volume
Methanol	3.75 mL
Chloroform	2.50 mL (added in two equal portions)
Water	1.25 mL
Total	≈6.25 mL

3. Divide equally into glass vials. Add chloroform and water (1:1) to each vial; vortex. Centrifuge 30 s at $2000 \times g$. Two phases form: lipids in the lower organic phase, aqueous debris above.
4. Evaporate aqueous phase under nitrogen stream. Collect lower organic fraction.

7.4 Tangential Flow Filtration (TFF)

In TFF, feed flow runs parallel (tangential) to the membrane surface rather than perpendicular. This shear flow prevents particle accumulation on the membrane. Particles and solvent passing perpendicularly through the membrane form the permeate; the retained fraction (retentate) is concentrated. TFF was used to concentrate and size-select MEVs prior to injection.



Figure 4: Tangential flow filtration apparatus.

Procedure: Flush the system with MilliQ; run sample through the membrane to retain MEVs of the desired size range. Sterilize the passage after use by recirculating sodium hydroxide and sodium hypochlorite solution.

7.5 Lipid Quantification (Phospho-Vanillin Assay)

A standard curve (absorbance vs. concentration) is constructed from serial dilutions of a known lipid stock, enabling determination of the concentration of the Bligh–Dyer extract.

7.5.1 Serial Dilution of Lipid Standard

Table 3: Serial dilution of lipid standard in chloroform.

Step	Volume taken (μL)	Chloroform added (μL)	Conc. (mg/mL)
Stock	–	–	10.00
1	20	20	5.00
2	20	20	2.50
3	20	20	1.25

7.5.2 Procedure

1. Add 180 μL concentrated H_2SO_4 to each 20 μL standard vial (total 200 μL). Heat at 100°C for 10 min; cool to room temperature. H_2SO_4 promotes sulfonation and breaks lipid esters into reactive carbonyl species.
2. Add 0.5 mL phospho-vanillin reagent. Incubate at 37°C for 15 min, then rest in dark for 45 min. Vanillin reacts with carbonyl breakdown products to form a pink complex.
3. Measure absorbance at 530–550 nm.

Table 4: Absorbance values obtained (triplicates per row, highest concentration first).

Absorbance (triplicates)		
3.500	3.500	3.500
3.500	3.500	3.500
1.610	1.699	2.299
0.987	0.830	0.944
0.276	0.258	0.255

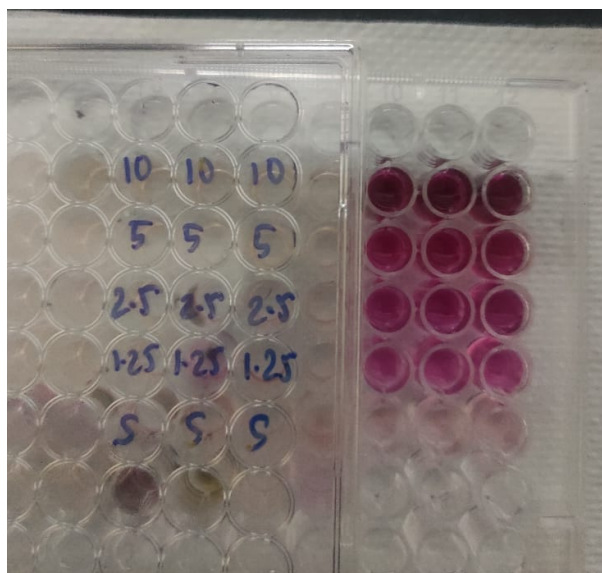


Figure 5: Colour gradient across lipid standard concentrations (96-well plate).

7.6 MEV Isolation

MEVs were isolated from *M. tuberculosis* H37Ra cultures by sequential centrifugation to remove cells and debris, followed by filtration and ultracentrifugation for purification.

1. Centrifuge culture at 5000 rpm, 15 min, $\times 3$; then 10,000 rpm, 10 min ($\times 1$). These steps pellet intact bacteria and large cell fragments.
2. Filter through 200 nm membrane to exclude particles larger than MEVs; apply TFF to concentrate and size-select the filtrate.
3. Centrifuge at $4000 \times g$, 15 min, then $15,000 \times g$, 15 min to remove residual aggregates.
4. Ultracentrifuge at $100,000 \times g$, 2 h to pellet and purify MEVs completely.

8 LUV Preparation

8.1 By Extrusion

Extrusion produces LUVs of controlled size by forcing a multilamellar vesicle suspension through a polycarbonate membrane of defined pore size.

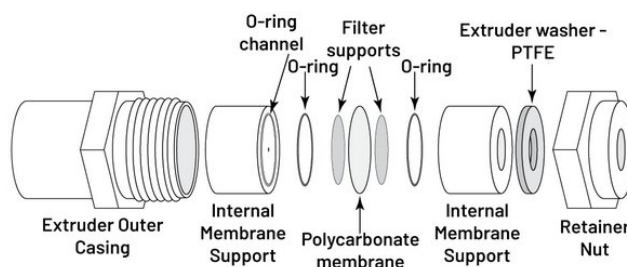


Figure 6: Schematic of the mini-extruder apparatus.

1. Soak all extruder parts (syringes, membrane support, washers, O-rings) in MilliQ for 1 h; dry thoroughly.
2. Wet filter supports with MilliQ using blunt forceps; place on internal membrane support (O-ring side). Lay polycarbonate membrane on one filter support; insert into outer casing and seal with retainer nuts.
3. Test for leaks: extrude 1 mL MilliQ and check for volume loss.
4. Load lipid mixture (bubble-free); extrude 20 times through the membrane. Collect in MCT; store on ice or at 4°C . Disassemble and soak components in MilliQ for >1 h after use.

8.2 LUV Preparation for Anisotropy Studies

LUVs were prepared from lipid mixtures of varying complexity to study how membrane composition affects IDP binding and anisotropy. The lipid systems used represent a compositional hierarchy: total brain extract (TBE) contains all brain lipids; brain phosphatidylcholine (BPC) and brain phosphatidylserine (BPS) are the dominant phospholipid classes; DOPC and DOPS are single defined lipids. This hierarchy allows systematic dissection of which lipid components drive IDP-membrane interactions.

8.2.1 Procedure

1. Aliquot lipid stocks (DOPS, DOPC, BPS, BPC, DOPC:DOPS 6:4, BPC:BPS 6:4, TBE, TBE- τ) into eight vials. Flush with nitrogen to prevent oxidation; place under vacuum.
2. Add buffer: PBS for TBE and TBE- τ vials; potassium phosphate buffer (480 μL MilliQ + 20 μL 5 M NaCl + 500 μL distributed across vials) for the rest.
3. Heat in water bath at 60°C for 20 min; vortex 4 min each to disrupt multilamellar layers. Chill on ice.
4. Transfer to MCTs; sonicate 6 min each. Verify size by DLS.
5. Prepare protein-LUV samples according to compositions in Tables 5 and 6.

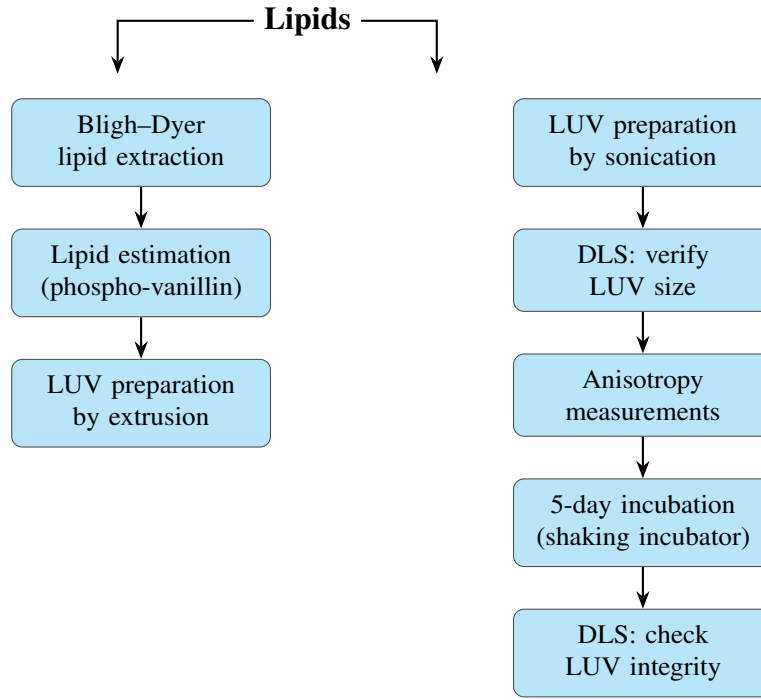


Figure 7: Parallel workflows for lipid processing and LUV preparation.

Table 5: Sample compositions for TBE, DOPC, DOPS, and DOPC:DOPS conditions (all volumes in μL).

Sample	LUV	A β 42	α -syn	KPi	NaCl	NaN ₃	MQ	Total
TBE	64	–	–	320	13	7	236.00	640
TBE + A β 42	64	14.5	–	320	13	7	221.50	640
TBE + α -syn	64	–	20.64	320	13	7	215.36	640
TBE + A β 42 + α -syn	64	14.5	20.64	320	13	7	200.86	640
DOPC	64	–	–	320	13	7	236.00	640
DOPC + A β 42	64	14.5	–	320	13	7	221.50	640
DOPC + α -syn	64	–	20.64	320	13	7	215.36	640
DOPC + A β 42 + α -syn	64	14.5	20.64	320	13	7	200.86	640
DOPS	64	–	–	320	13	7	236.00	640
DOPS + A β 42	64	14.5	–	320	13	7	221.50	640
DOPS + α -syn	64	–	20.64	320	13	7	215.36	640
DOPS + A β 42 + α -syn	64	14.5	20.64	320	13	7	200.86	640
DOPC:DOPS	64	–	–	320	13	7	236.00	640
DOPC:DOPS + A β 42	64	14.5	–	320	13	7	221.50	640
DOPC:DOPS + α -syn	64	–	20.64	320	13	7	215.36	640
DOPC:DOPS + A β 42 + α -syn	64	14.5	20.64	320	13	7	200.86	640

9 SDS-PAGE

9.1 Principle

SDS-PAGE (Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis) separates proteins by molecular weight. SDS denatures proteins and confers uniform negative charge proportional to chain length, eliminating shape and charge differences. Under an electric field, proteins migrate through the polyacrylamide matrix toward the anode; smaller proteins travel faster. A two-gel system is used: a 5% stacking

Table 6: Sample compositions for BPC, BPS, and BPC:BPS with τ and heparin (all volumes in μL).

Sample	LUV	τ	Heparin	KPi	NaCl	NaN_3	MQ	Total
BPC	64	–	–	320	13	7	236	640
BPC + τ + Heparin	64	80	80	320	13	7	76	640
BPS	64	–	–	320	13	7	236	640
BPS + τ + Heparin	64	80	80	320	13	7	76	640
BPC:BPS	64	–	–	320	13	7	236	640
BPC:BPS + τ + Heparin	64	80	80	320	13	7	76	640

gel (pH 6.8) concentrates all proteins into a tight starting band; a 15% resolving gel (pH 8.8) separates them by size.

Gel Compositions

Table 7: Composition of 15% resolving gel (10 mL) and 5% stacking gel (5 mL).

Component	Stock	Resolving (mL)	Stacking (mL)	Function
Acrylamide/Bis (29:1)	30%	5.00	0.83	Gel matrix and crosslinker
Tris-HCl	1.5 M pH 8.8 / 1.0 M pH 6.8	2.50	0.63	pH control
SDS	10%	0.10	0.05	Denaturation; uniform charge
Distilled water	–	2.30	3.40	Volume adjustment
APS	10%	0.10	0.05	Free-radical initiator
TEMED	Pure	0.01	0.0025	Polymerization catalyst

9.2 Procedure

1. Assemble clean glass plates in casting stand; seal bottom. Prepare resolving gel; pour between plates leaving stacking-gel space. Overlay with isopropanol to flatten the surface and prevent oxygen inhibition. Polymerize 30–45 min.
2. Remove isopropanol; prepare and pour stacking gel; insert comb immediately. Polymerize 20–30 min.
3. Remove comb; place assembly in electrophoresis tank; fill chambers with Tris-glycine-SDS running buffer. Load pre-boiled samples and molecular weight marker.
4. Run at 80 V through stacking gel; increase to 120 V through resolving gel until dye front reaches the bottom.
5. Stain with Coomassie Brilliant Blue; destain to visualize bands.

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